

Alaedine Benani MD, MSc

Clinical Machine Learning | Cardiovascular Prevention

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Research Focus

Physician and PhD candidate working on machine learning for cardiovascular prevention and clinical decision support. Research integrates multimodal modeling, synthetic clinical data and evidence synthesis, with particular attention to model interpretability, prescribing variability and rigorous evaluation.

Education and Clinical Training

PhD candidate, Machine Learning for Clinical Decision Support

2024–present

Sorbonne Université, LIMICS, Inserm, Paris, France. Expected completion: May 2027.

Research with Prof. Xavier Tannier, in partnership with Zoī and AP-HP.

Doctor of Medicine (MD)

June 2026

Medical training at Université Toulouse III – Paul Sabatier and Université Paris Cité.

Dual medicine-science curriculum, including the Inserm Liliane Bettencourt School.

Residency in Vascular Medicine

2022–present

Université Paris Cité; Assistance Publique – Hôpitaux de Paris (AP-HP).

Clinical rotations (November 2022–May 2024) in cardiovascular rehabilitation, inpatient care, vascular investigations and day hospital; research leave since May 2024.

Master 2 (MSc), Biomedical Engineering

2018–2019

Université Paris-Saclay / École Polytechnique / HEC Paris, France.

Training in machine learning, technological innovation and entrepreneurship; research internship at Owkin.

Research and Professional Experience

Machine Learning Researcher and Doctoral Researcher

May 2024–present

LIMICS (Sorbonne Université / Inserm) and Zoī, Paris; CIFRE doctoral partnership.

- Model cardiovascular prevention prescriptions using real, synthetic and hybrid clinical datasets; first-author study with cross-validation and held-out evaluation.
- Lead a systematic review comparing multimodal and unimodal models across 97 studies and 12 specialties, including PROBAST risk-of-bias assessment (preprint).
- Study within- and between-physician prescribing variability and its implications for model evaluation.
- Contribute to the Zoī prospective cohort: published profile of the first 1,000 participants, with 196 biomarkers, over 500 self-reported items and multimodal imaging.

Medical Project Manager / Data Consultant

April 2020–February 2024

Synapse Medicine and independent collaborations, including Raidium. Contributed to Goodmed, Synapse Pro, the Synapse Copilot knowledge base and medical question-answering research (ClinicalNLP 2024).

Data Scientist Intern

July–September 2019

Owkin, Paris, France. Master's research internship in machine learning for CT image analysis within the radiology team.

Publications

Name in bold. Articles published in French are marked.

Journal Articles and Reviews

1. Lepoittevin M, Bauvin P, **Benani A**, Attias P, Steg PG, Vidal-Petiot E, Bodard S. Impact of the new 2024 ESC blood pressure classification in a low-risk cohort: the 70 mmHg diastolic threshold may reduce risk discrimination in primary prevention. *Journal of Hypertension*. Published online September 2, 2026. doi: 10.1097/HJH.0000000000004434.
2. Balia M, Zenasni F, Lepoittevin M, **Benani A**, Bianchi R, Julian A, Bodard S, Bringer M. Biological Markers of Burnout Severity: A Systematic Review. *Stress and Health*. 2026;42(4):e70205. doi: 10.1002/smi.70205.
3. Bauvin P, **Benani A**, Lepoittevin M, Sentilhes M, Bringer M, Lecheheb DA, De Luca N, Ohayon S, Dalle C, Tannier X, Steg PG, Bodard S. Zoï cohort, a prospective cohort with comprehensive phenotyping for preventive medicine in France, first 1000 participants: cohort profile. *BMJ Open*. 2026;16(5):e111119. doi: 10.1136/bmjopen-2025-111119.
4. Bodard S, Segui P, Poirier F, Bauvin P, Lepoittevin M, **Benani A**, Mounier M, Herpe G, Tacher V. Feasibility of Latest-Generation Cone-Beam CT for Thoracic Imaging: A Pilot Study. *Academic Radiology*. 2026;33(5):2257–2269. doi: 10.1016/j.acra.2025.12.029.
5. Bodard S, **Benani A**, Lepoittevin M, Vibert J, Cornelis FH. Robotics in oncologic interventional radiology: from technological innovation to personalized medicine. *Médecine/Sciences*. 2026;42(2):201–204. [In French.] doi: 10.1051/medsci/2026002.
6. Lepoittevin M, Greig J, Erol O, **Benani A**, Bauvin P, Azzolini C, Donati S, Dubois B, Boscher C, Bodard S. Retinal biomarkers for early Alzheimer's detection: a systematic review of optical coherence tomography (OCT) findings. *BMJ Open Ophthalmology*. 2026;11(1):e002328. doi: 10.1136/bmjophth-2025-002328.
7. **Benani A**, Vibert J, Demuth S. Synthetic data in medicine: Generation, evaluation and limits. *Médecine/Sciences*. 2024;40(8–9):661–664. [In French.] doi: 10.1051/medsci/2024091.

Conference Proceedings

8. **Benani A**, Grosgeorge D, Bauvin P, Ohayon S, Bodard S, Messas E, Tannier X. Machine Learning for Cardiovascular Prevention Prescriptions: Real-World vs. Synthetic Data. *Studies in Health Technology and Informatics*. 2026;336:363–367. doi: 10.3233/SHTI260178.
9. Khlaut J, Dancette C, Ferreres E, **Bennani A**, Hérent P, Manceron P. Efficient Medical Question Answering with Knowledge-Augmented Question Generation. In: *Proceedings of the 6th Clinical Natural Language Processing Workshop*. Association for Computational Linguistics; 2024:10–20. doi: 10.18653/v1/2024.clinicalnlp-1.2.

Publications (continued)

Scholarly Perspectives and Scientific Commentary

10. **Benani A**, Messas E. Nobel Prize in physics 2024: John J. Hopfield and Geoffrey E. Hinton. From Hopfield and Hinton to AlphaFold: The 2024 Nobel Prize honors the pioneers of deep learning. *Médecine/Sciences*. 2025;41(3):277–280. [In French.] doi: 10.1051/medsci/2025036.
11. **Benani A**. History and vision of artificial intelligence. *Médecine/Sciences*. 2024;40(3):283–286. [In French.] doi: 10.1051/medsci/2024022.
12. **Benani A**, Gaudet C, Varennes O, Vellas D. Scientific news selected by AMPS students [L'actualité scientifique vue par les étudiants de l'AMPS]. *Médecine/Sciences*. 2018;34(4):313. [In French.] doi: 10.1051/medsci/20183404011.

Preprints (not peer reviewed)

13. Lepoittevin M, Bringer M, Balia M, Bauvin P, **Benani A**, Zenasni F, Bodard S. Latent chronic stress: identification and characterization in a preventive medicine cohort. *medRxiv*. 2025. doi: 10.1101/2025.11.21.25340737.
14. Etienne L, Bauvin P, **Benani A**, Lepoittevin M, Bodard S. Advancing cardiovascular disease risk prediction beyond conventional methods: a systematic review of multimodal machine learning models integrating traditional clinical factors and multi-omics data. *medRxiv*. 2025. doi: 10.1101/2025.10.07.25337473.
15. **Benani A**, Ohayon S, Laleye F, Bauvin P, Messas E, Bodard S, Tannier X. Is Multimodal Better? A Systematic Review of Multimodal versus Unimodal Machine Learning in Clinical Decision-Making. *medRxiv*. 2025. doi: 10.1101/2025.03.12.25322656.

Selected Scientific Presentations

February 2026 Intra- and inter-physician variability as a barrier to robust machine learning prediction.

Poster presentation, Sorbonne Université Doctoral School ED 393 Conference, Saint-Malo, France.

January 2026 French Cardiology Congress.

Oral presentation, France.

October 2025 French Society of Vascular Medicine (SFMV) Congress.

Oral presentation, France.

July 2025 Longevity Summit Dublin.

Oral presentation, Dublin, Ireland.

June 2025 Artificial Intelligence in Medicine (AIME) Conference.

Oral presentation.

February 2025 State of the art of artificial intelligence in vascular medicine.

Departmental presentation, Hôpital Européen Georges-Pompidou, AP-HP, Paris, France.

January 2025 Introduction to synthetic data.

Research seminar, LIMICS, Inserm, Paris, France.

Teaching and Educational Development

Continuing medical education, Médéré. Expert instructor for an 8-hour course on artificial intelligence in healthcare for physicians, offered in person and online.

2025: AI workshops and university teaching. Workshops at the SFMV Congress (October) and the regional vascular medicine association in Reims (September); lecture on AI in preventive medicine for the DUMAA university diploma (April).

Co-founder, Freecn.io

June 2020–January 2024

Developed a nonprofit medical education platform for preparation for the French national residency examination. Built a team of more than 25 volunteers; the platform served more than 15,000 medical students.

Project Manager Trainee, Centre de Recherches Interdisciplinaires

April–September 2019

Developed machine learning training initiatives for healthcare professionals with Olivier Bory and Cloé Bami, Paris, France.

Academic and Professional Service

Resident Representative, Medical Executive Committee

January 2024–present

AP-HP Paris Centre University Hospital Group. Represent medical residents across seven hospitals, including Hôpital Européen Georges-Pompidou.

Editorial and peer-review service. Editorial board member, *Médecine/Sciences*; reviewer, *European Journal of Preventive Cardiology*.

Leadership in Physician-Scientist Associations

2017–2019

Board member, European MD/PhD Association (November 2017–January 2019); central bureau member, Association Nationale des Doubles Coursus en Santé (June 2017–June 2019).

Research Methods and Technical Skills

Programming: Python, R, MATLAB; version control and model deployment.

Machine learning: regression, random forests, gradient boosting / XGBoost, clustering, neural networks and transformers; synthetic data generation and evaluation.

Clinical data analysis: data integration and cleaning, feature engineering, missing-data handling, visualization, cross-validation and held-out model evaluation.

Evidence synthesis: systematic reviews, multimodal model comparison and PROBAST risk-of-bias assessment.

Languages: French (native); English (professional working proficiency).

Additional Training

2025: VASCERN Summer School on Rare Vascular Diseases.

2017–2018: Certificate in Political Science, Sciences Po Toulouse.

Academic References

Prof. Xavier Tannier. Professor, Sorbonne Université; LIMICS, Inserm.

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Prof. Emmanuel Messas. Head of Vascular Medicine, Hôpital Européen Georges-Pompidou, AP-HP.

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